**Python Introduction and Basics:-**

Python is a high-level, interpreted, and general-purpose programming language created by Guido van Rossum and released in 1991.

**It is widely used for:**

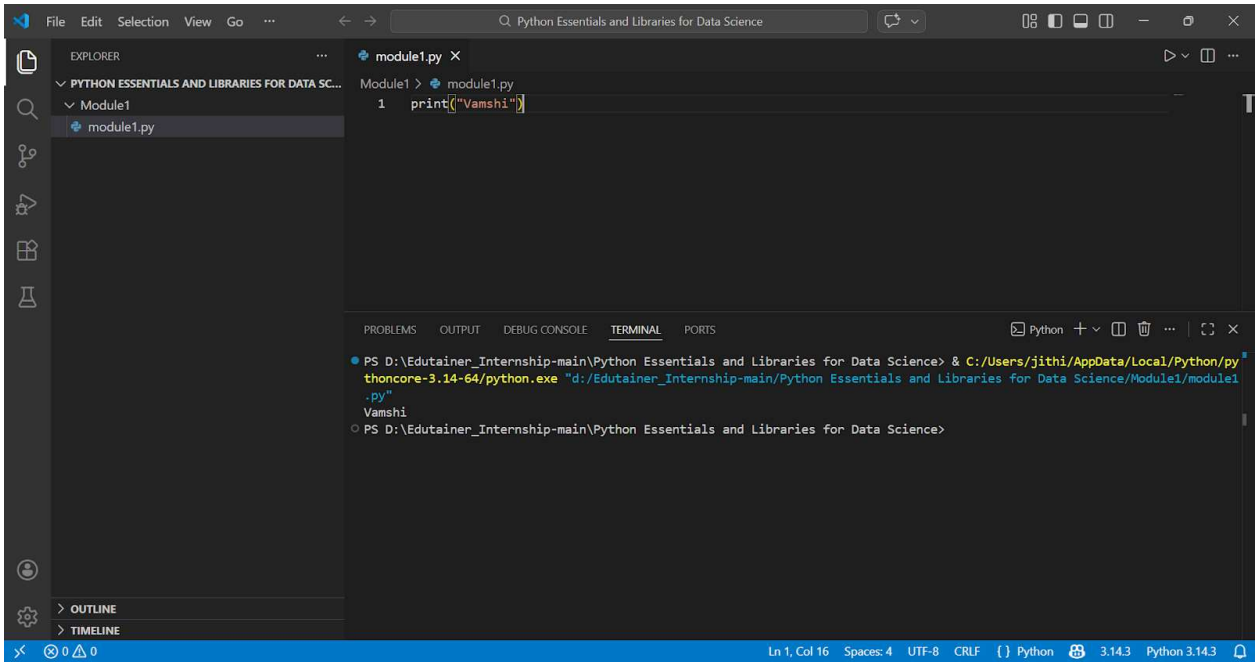
- Web development
- Data Science
- Artificial Intelligence
- Machine Learning
- Automation
- Software development

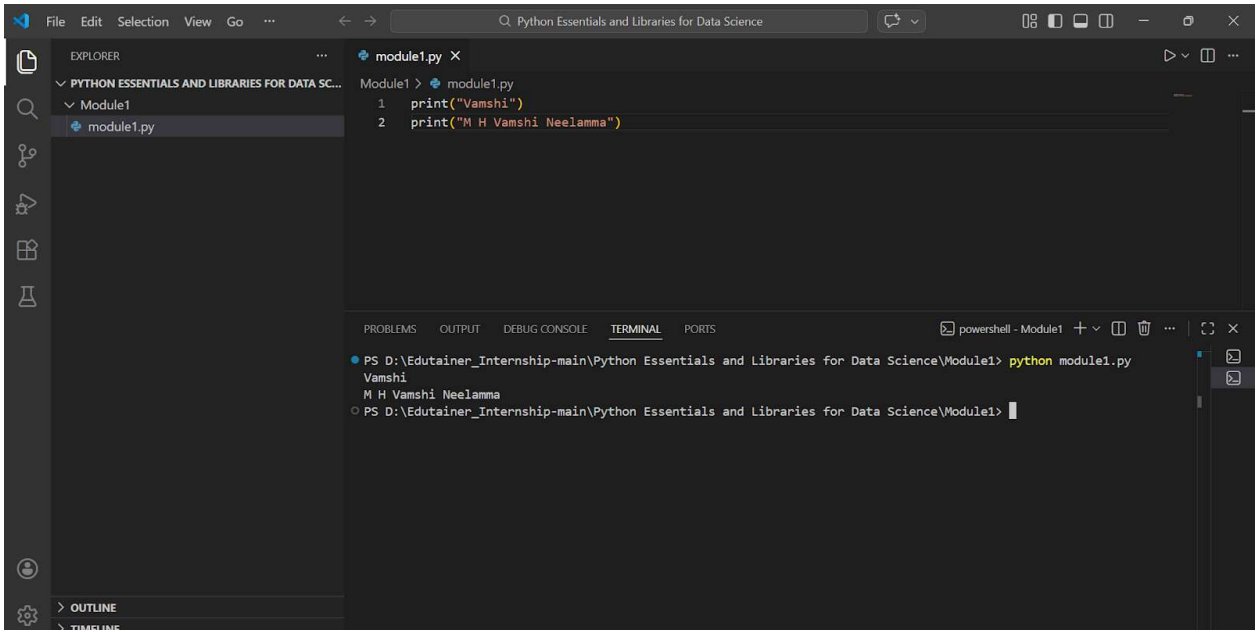
**Python Basics:-**

Python syntax is simple compared to other languages.

**To print something ,the syntax is:-**

print("Vamshi")

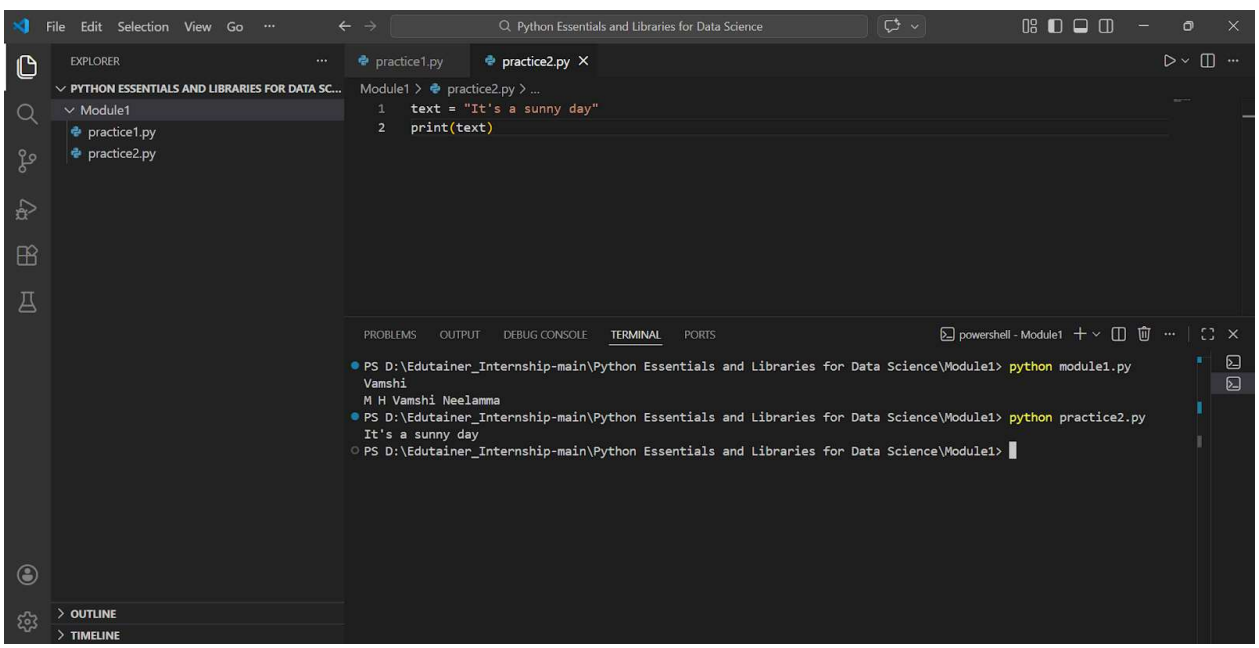




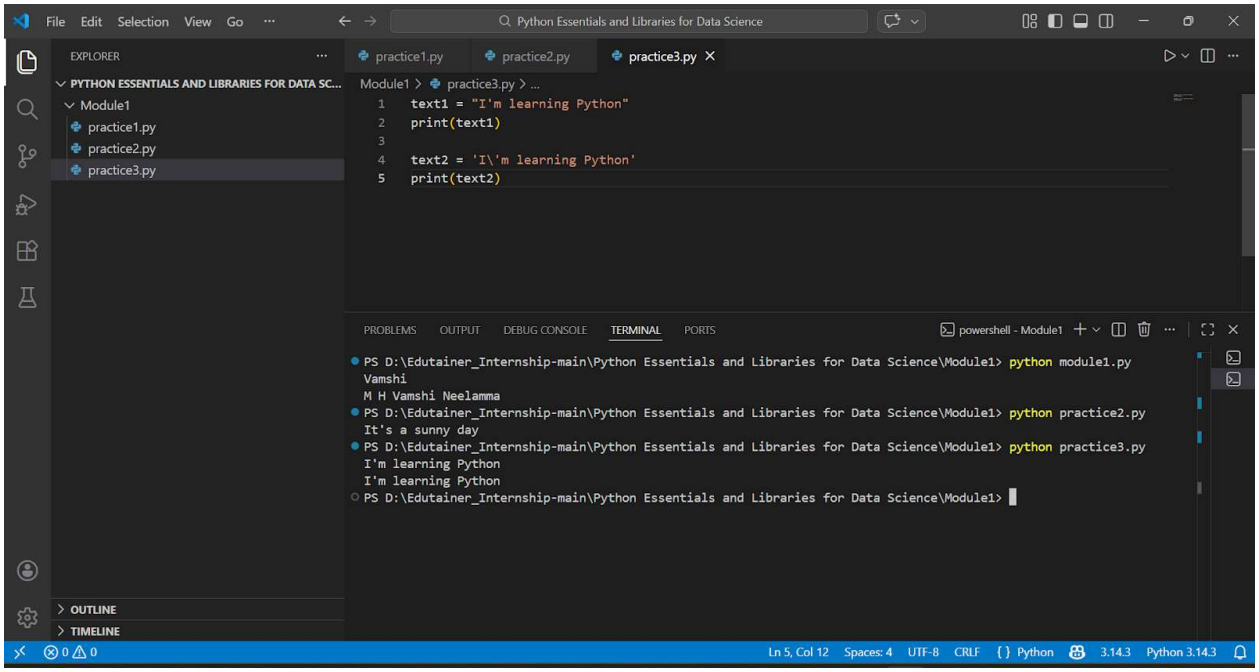

There is no necessity of using double quotes(“....”),instead we can use single quotes also (‘.....’)

**But where should we use single quotes?**

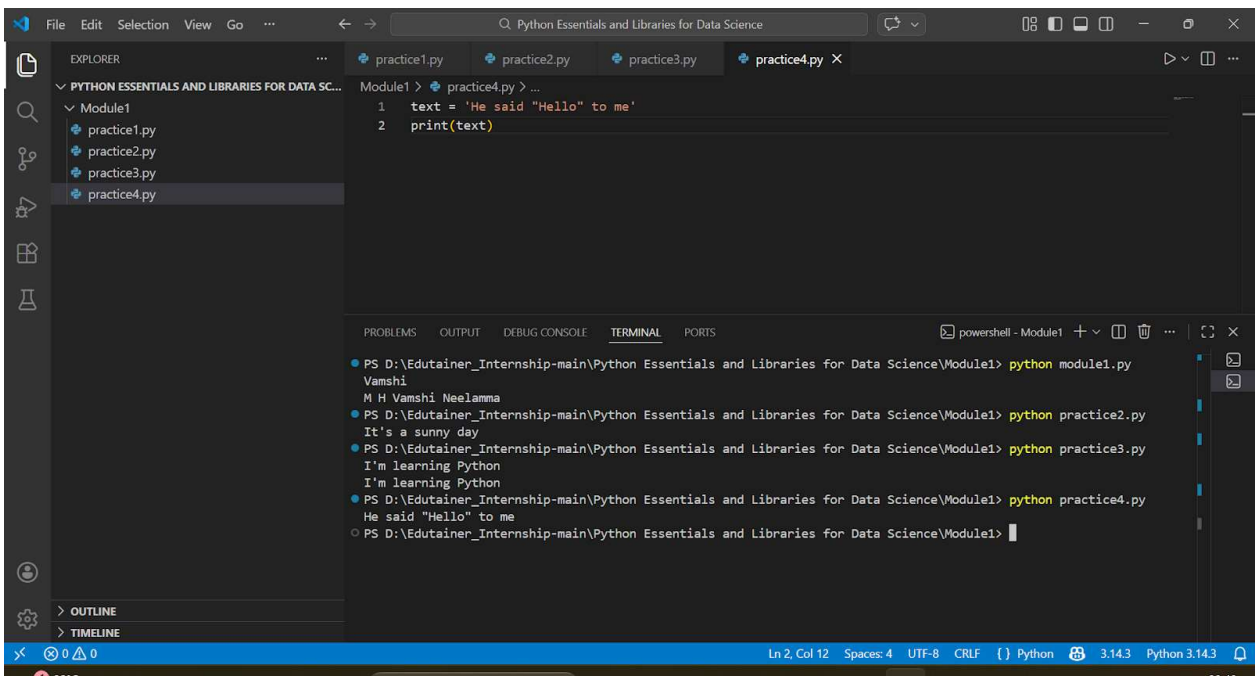
- Use double quotes if the text contains a single quote (')




- Double quotes help avoid using escape characters ().If you use single quotes, you must escape:




- If the sentence contains double quotes, then use single quotes.




**Types of data:-**

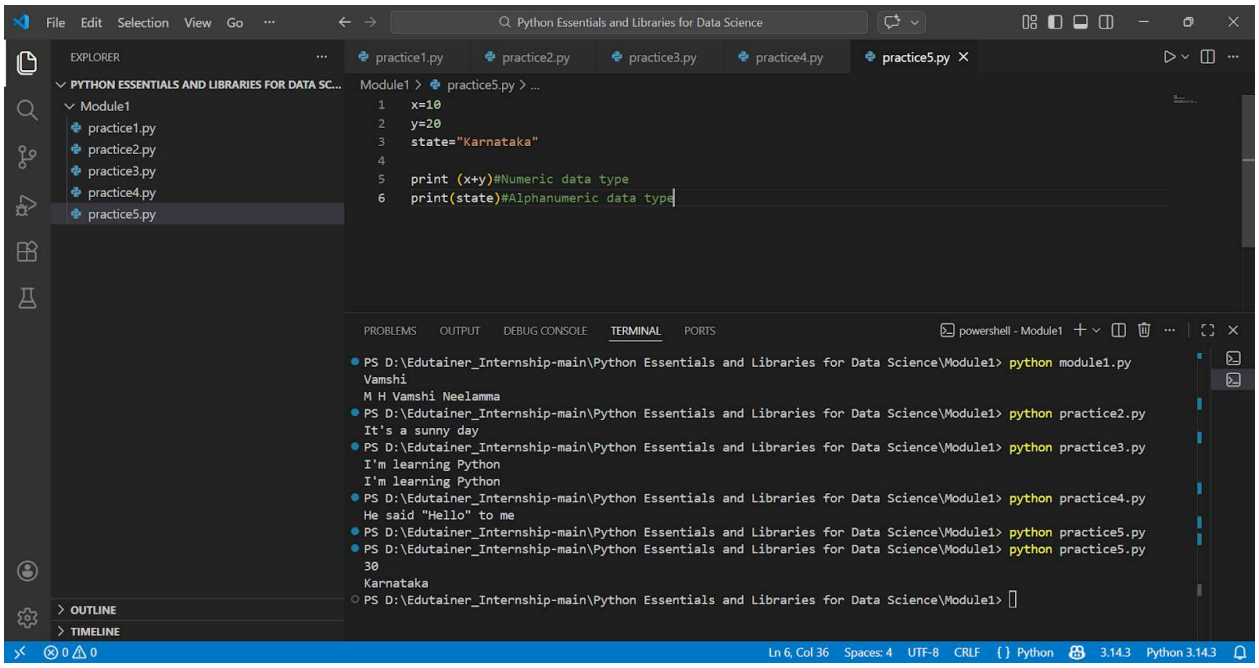
**1.Numeric data**

Numeric data refers to data that consists of numbers and can be used for mathematical calculations such as addition, subtraction, multiplication, and division.

Ex:-int,float,complex

**2.Alphanumeric data:-**

Alphanumeric data contains a combination of letters (A–Z, a–z) and numbers (0–9).
In Python, this type of data is stored as a string.




**Python Installation:-**

**First install python software.**

1. Open your browser.
2. Search “python download” and open the official Python website.
3. Click Download Python (latest version).
4. Open the downloaded file (example: python-3.xx.x-amd64.exe).
5. VERY IMPORTANT: On the first screen, tick “Add python.exe to PATH”
6. Click Install Now.
7. Wait until it finishes and click Close.
8. To verify the installation
   ->Go to command prompt
   ->Type python  --version
   ->If you get the version of python,its installed correctly
   ->Also check the python package installer using pip –version

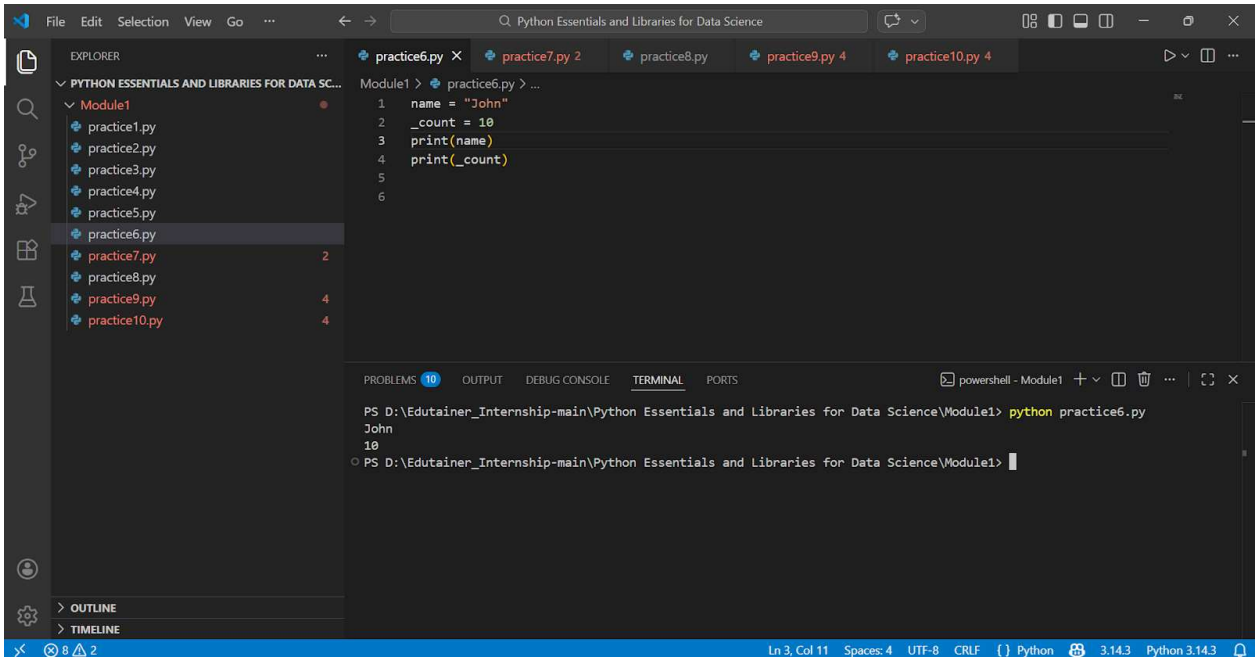
Then install vscode.

**Variable Naming Conventions:-**

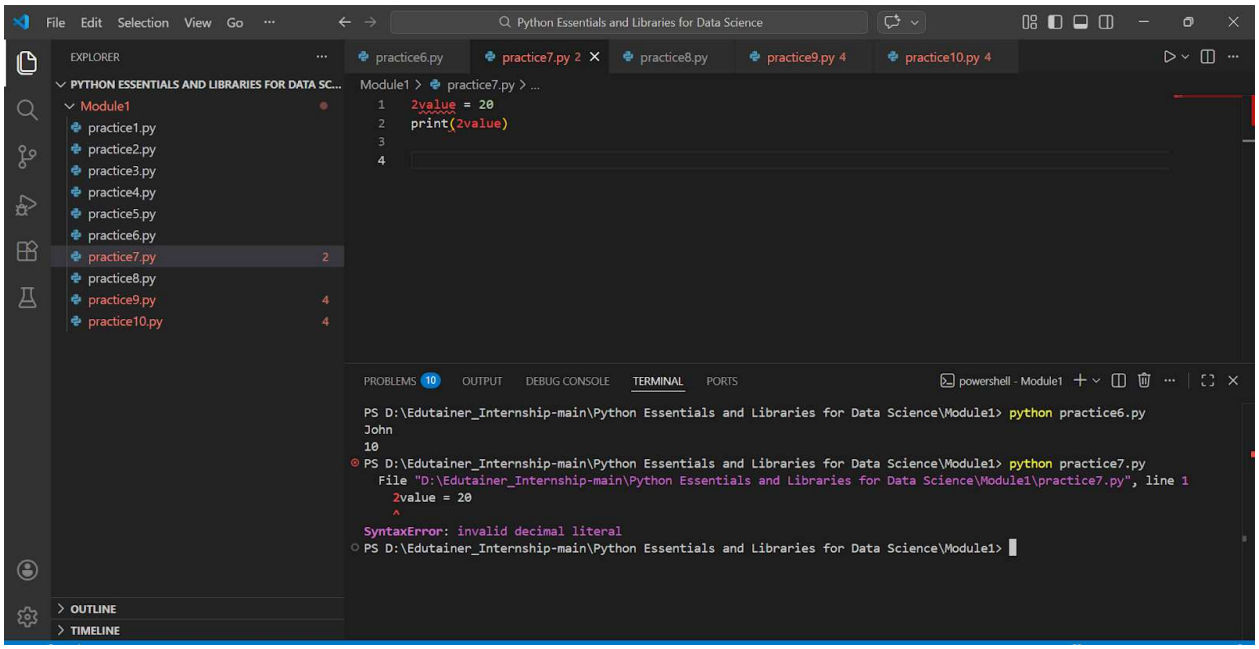
Variable naming conventions are rules and best practices used to give meaningful names to variables so that code becomes easy to read and understand.

Basic rule for naming variables:-

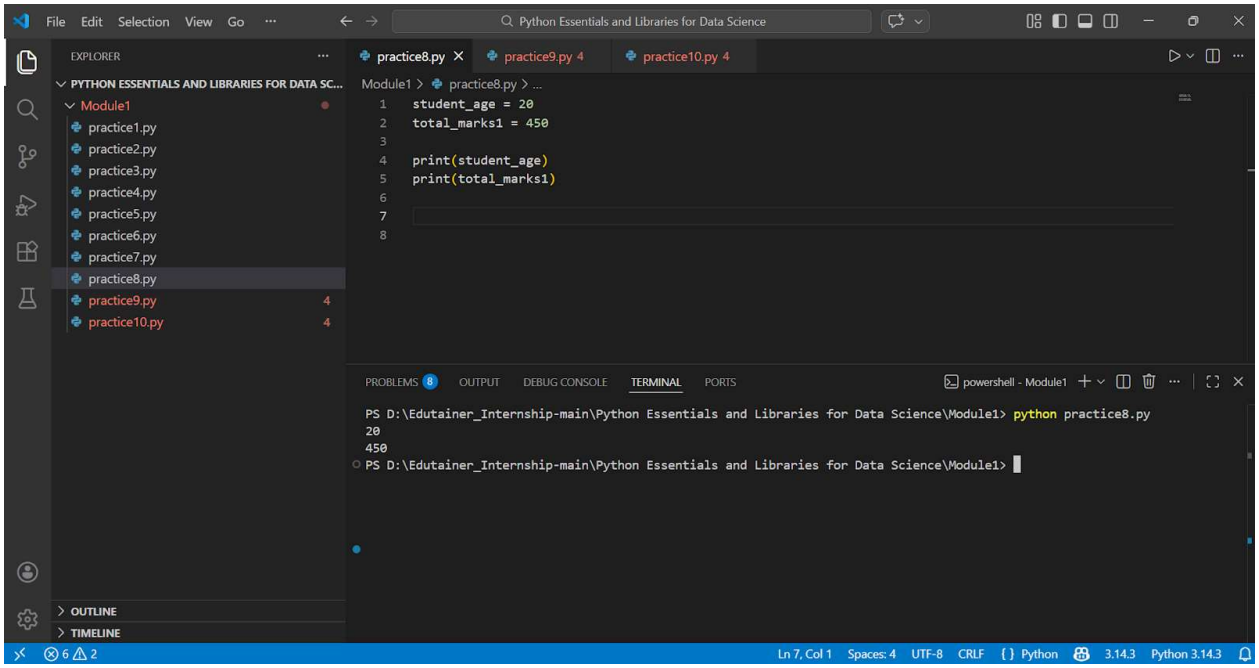
1.A variable name must start with a letter (a–z, A–Z) or underscore (_).



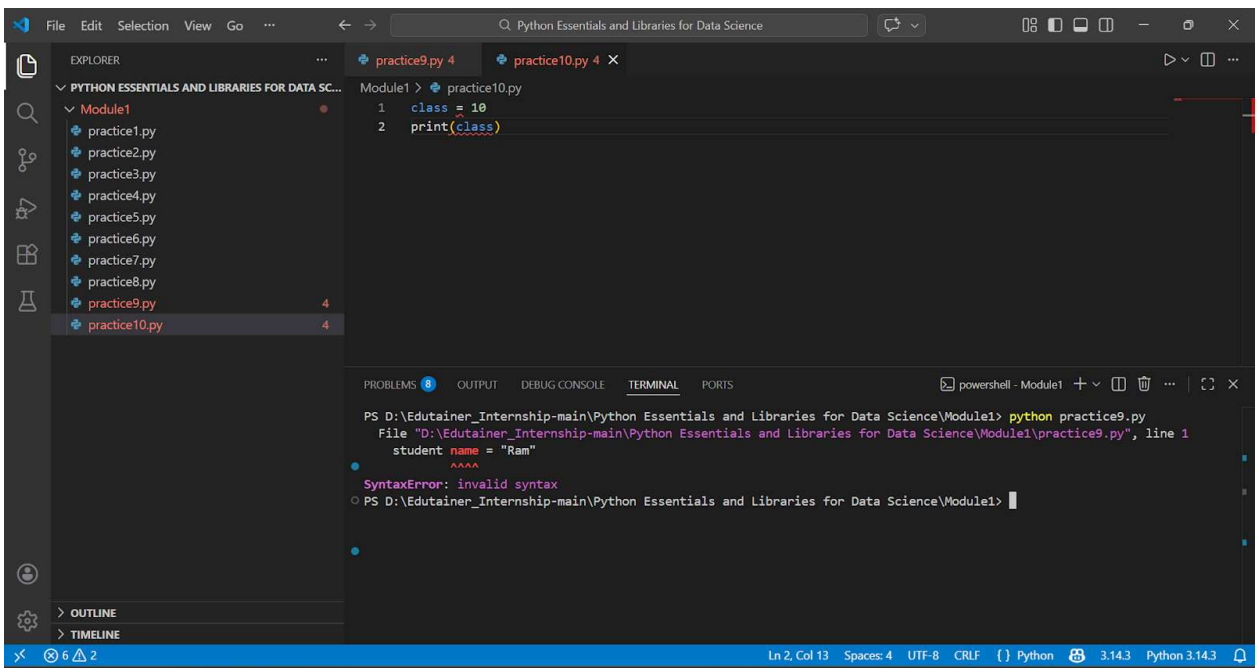

2.A variable name cannot start with a number.



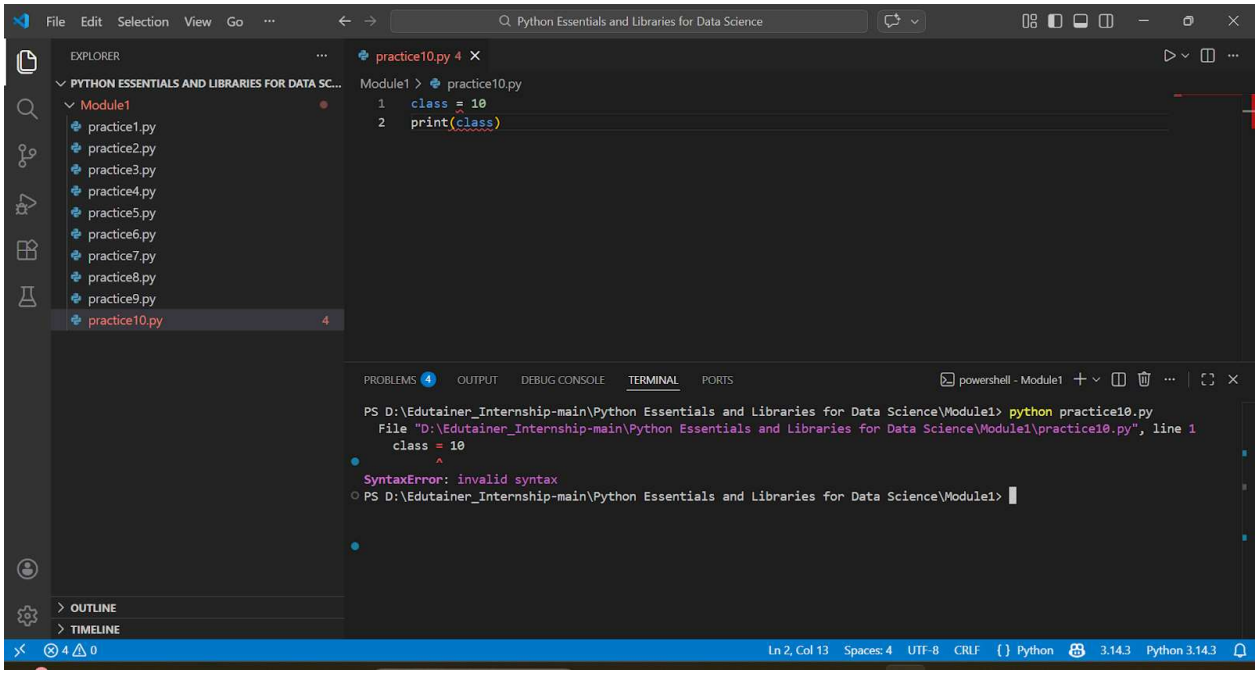

3.Variable names can contain letters, numbers, and underscores.




4.Spaces are not allowed in variable names.

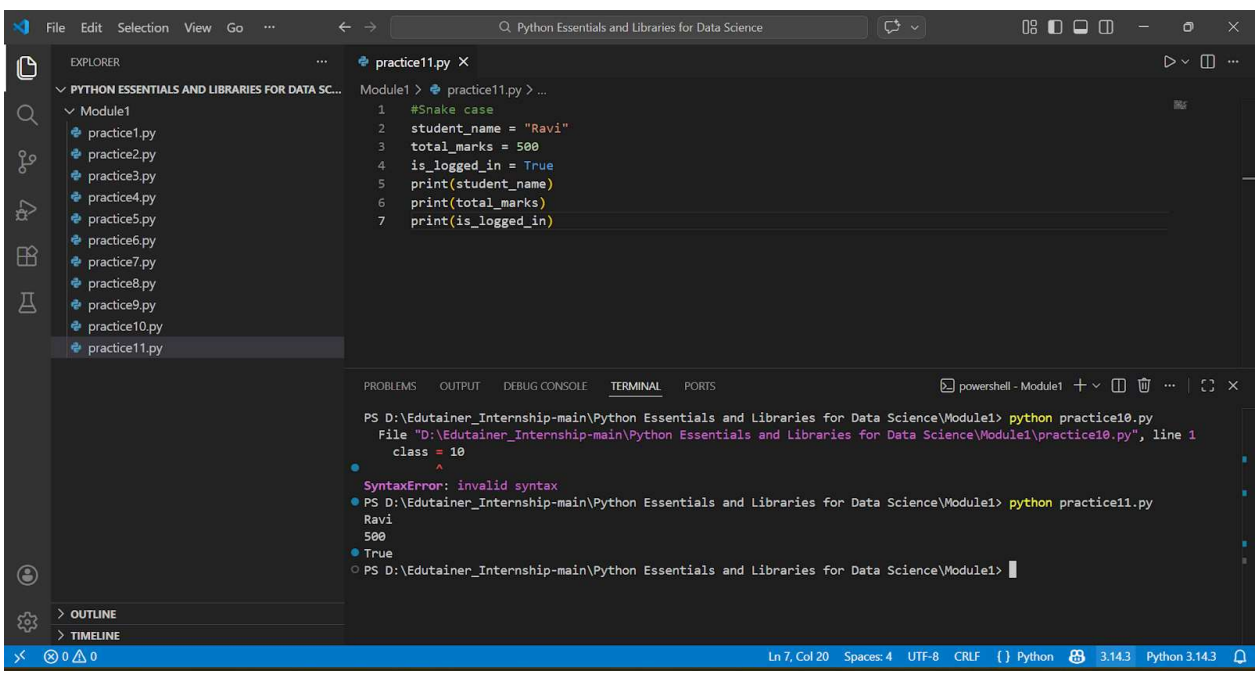



5.Python keywords cannot be used as variable names.

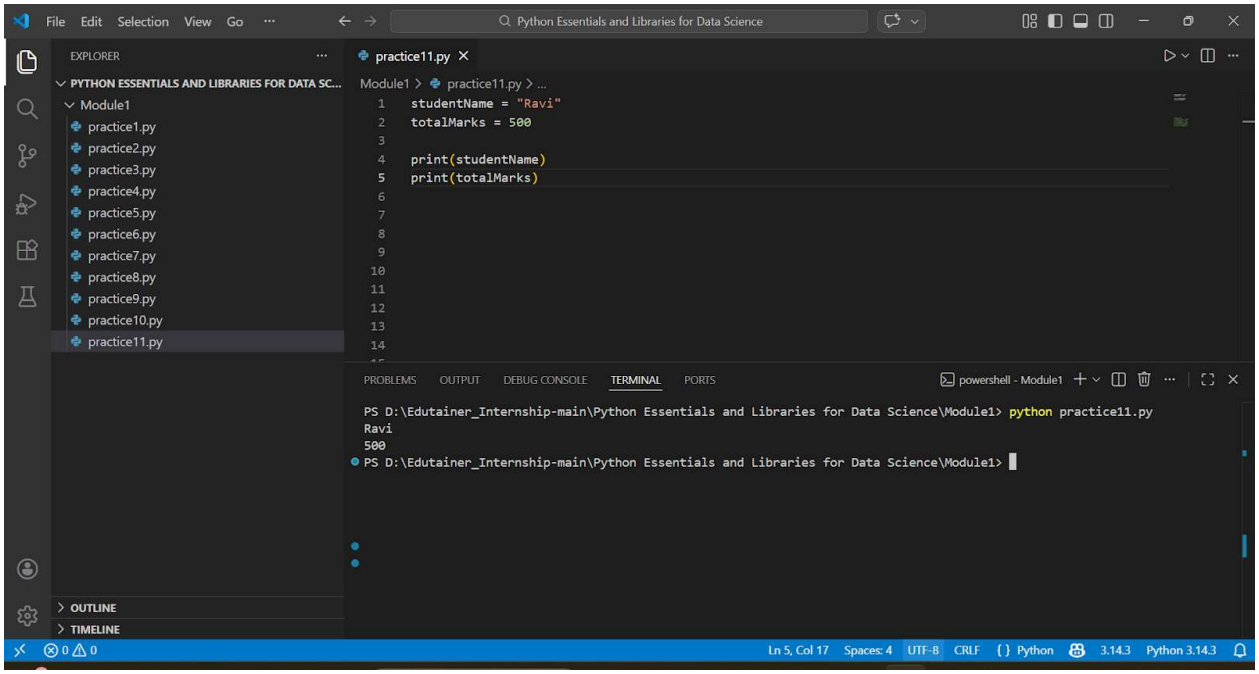



**Python variable naming style:-**

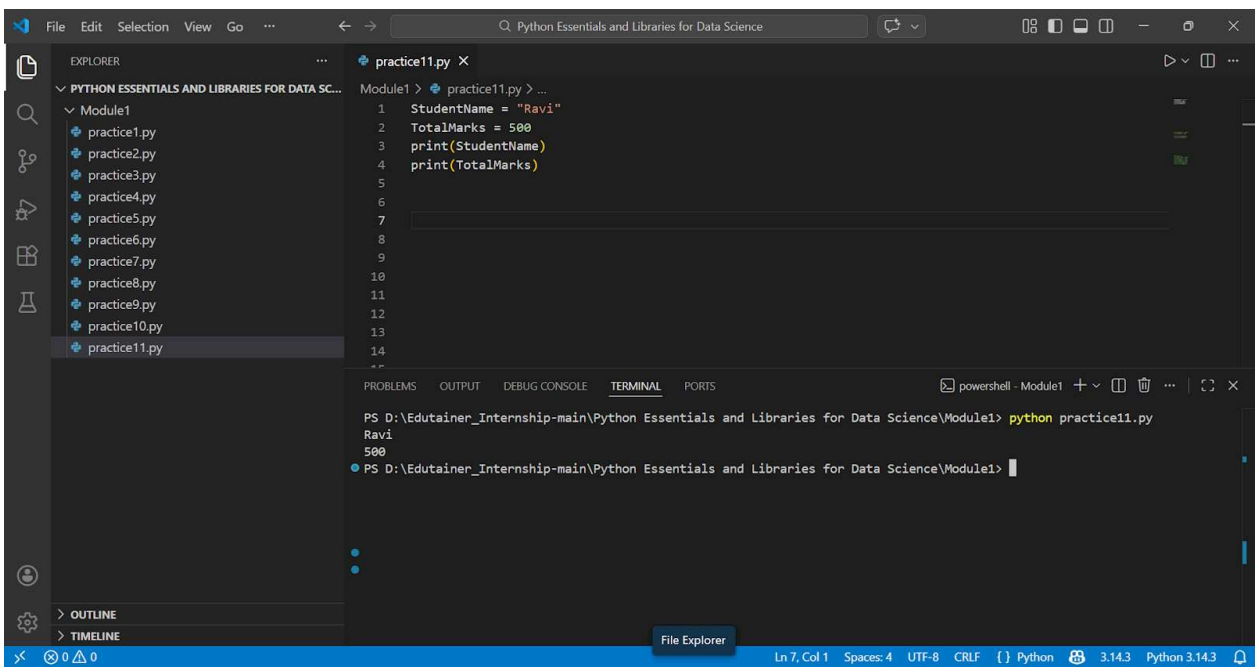
1.Snake Case




2.Camel Case




3.Pascal Case




**Data types in python:-**

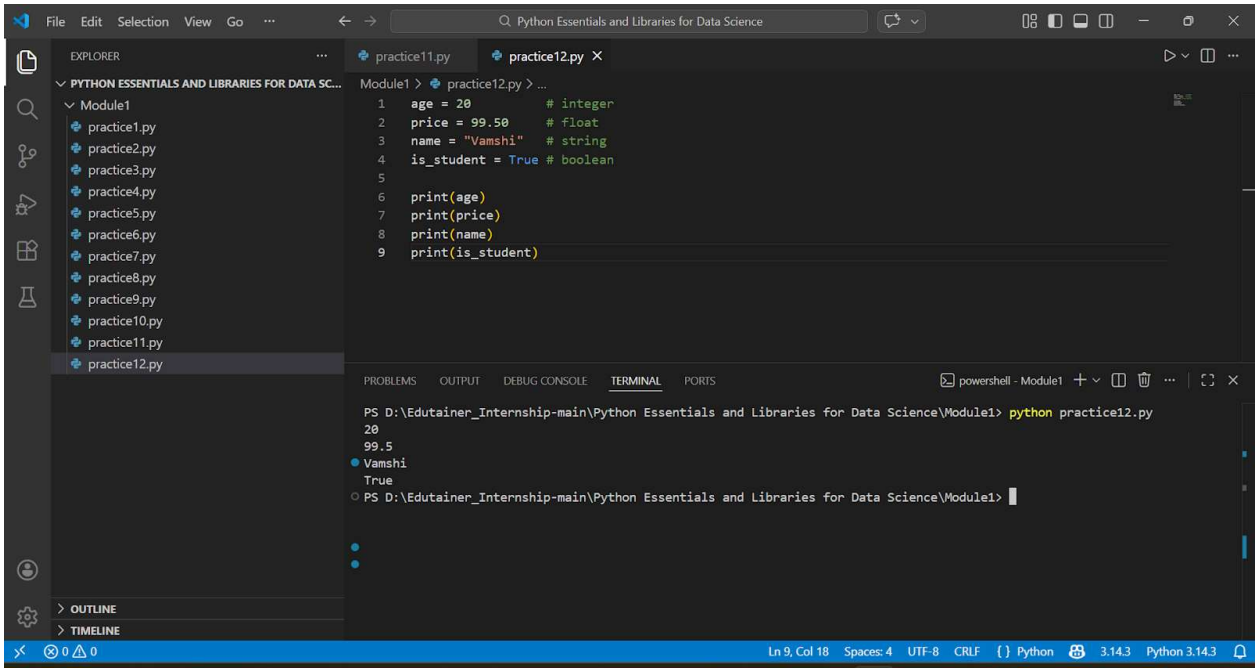
A data type defines the type of value stored in a variable.

**Common data types are:-**

int-integer numbers

float-decimal numbers

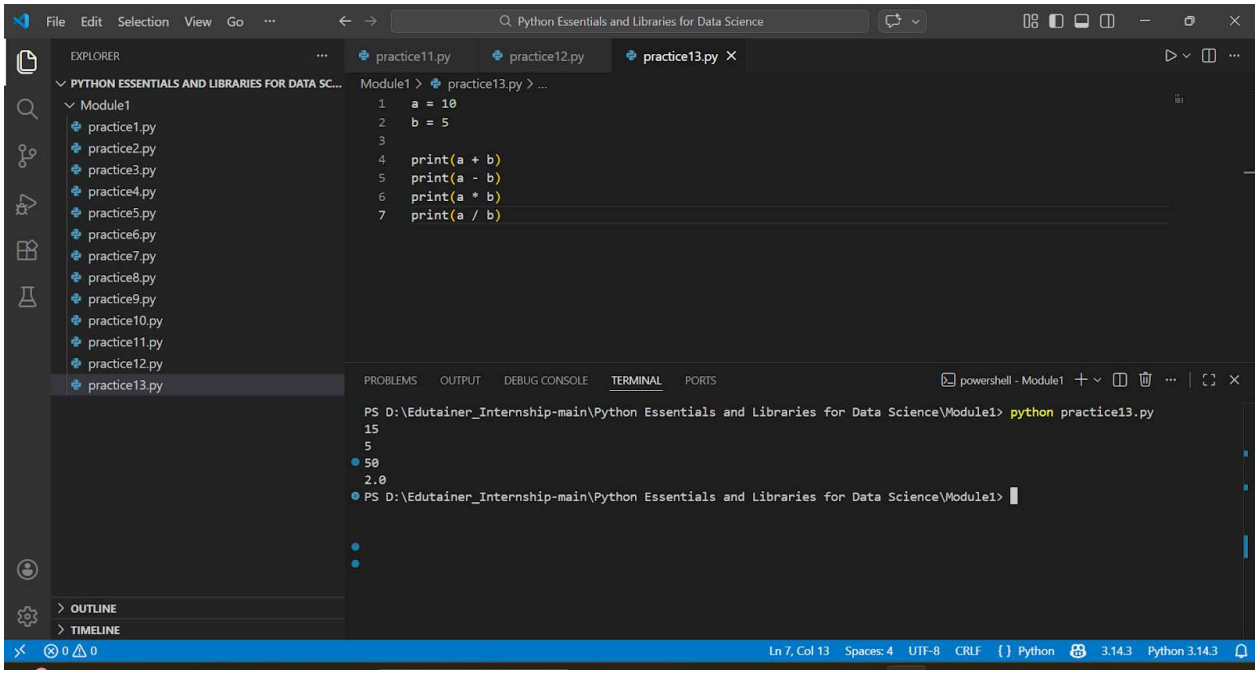
str-text
Bool-true or false values




**Arithmetic Operators :-**

Arithmetic operators are used to perform mathematical operations.

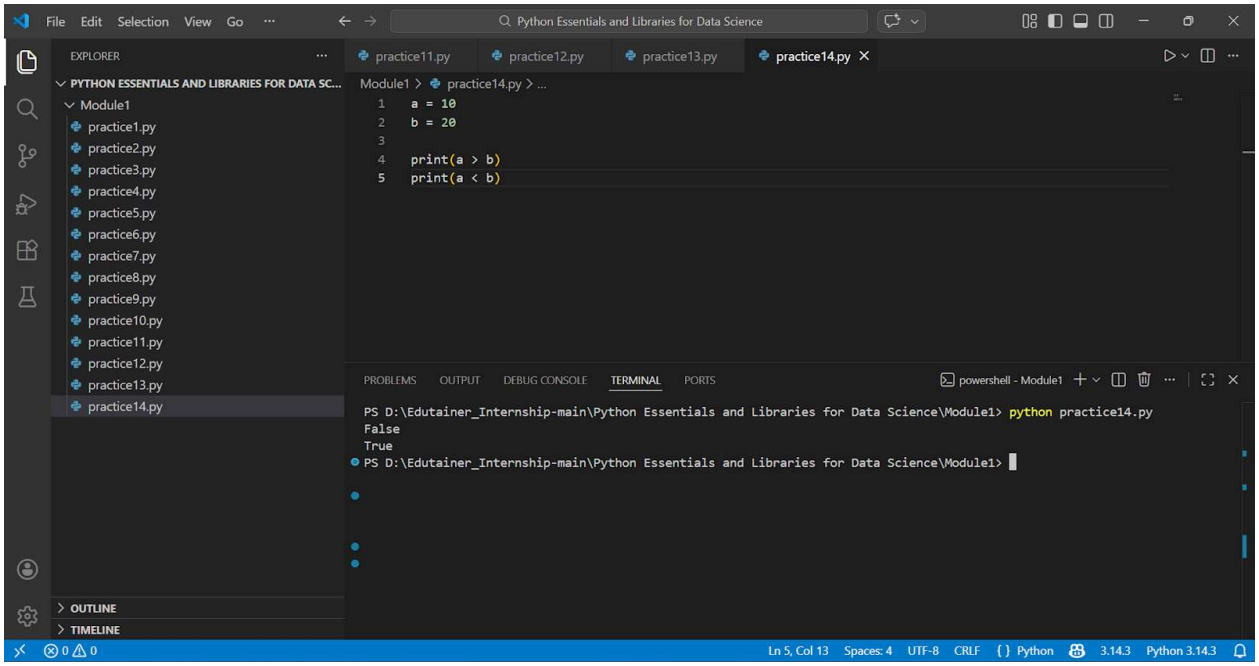
| Operator | Meaning | Example |
|---|---|---|
| + | Addition | 5 + 3 |
| - | Subtraction | 5 - 3 |
| * | Multiplication | 5 * 3 |
| / | Division | 5 / 3 |
| % | Modulus (remainder) | 5 % 3 |
| // | Floor division | 5 // 3 |
| ** | Power | 2 ** 3 |




**Comparison operator:-**

Comparison operators compare two values and return True or False.

| Operator | Meaning | Example |
|---|---|---|
| == | Equal | a == b |
| != | Not equal | a != b |
| > | Greater than | a > b |
| < | Less than | a < b |
| >= | Greater or equal | a >= b |
| <= | Less or equal | a <= b |




**Program to check if a number is even or odd.**

A number is even if it is divisible by 2.

A number is odd if it is not divisible by 2.

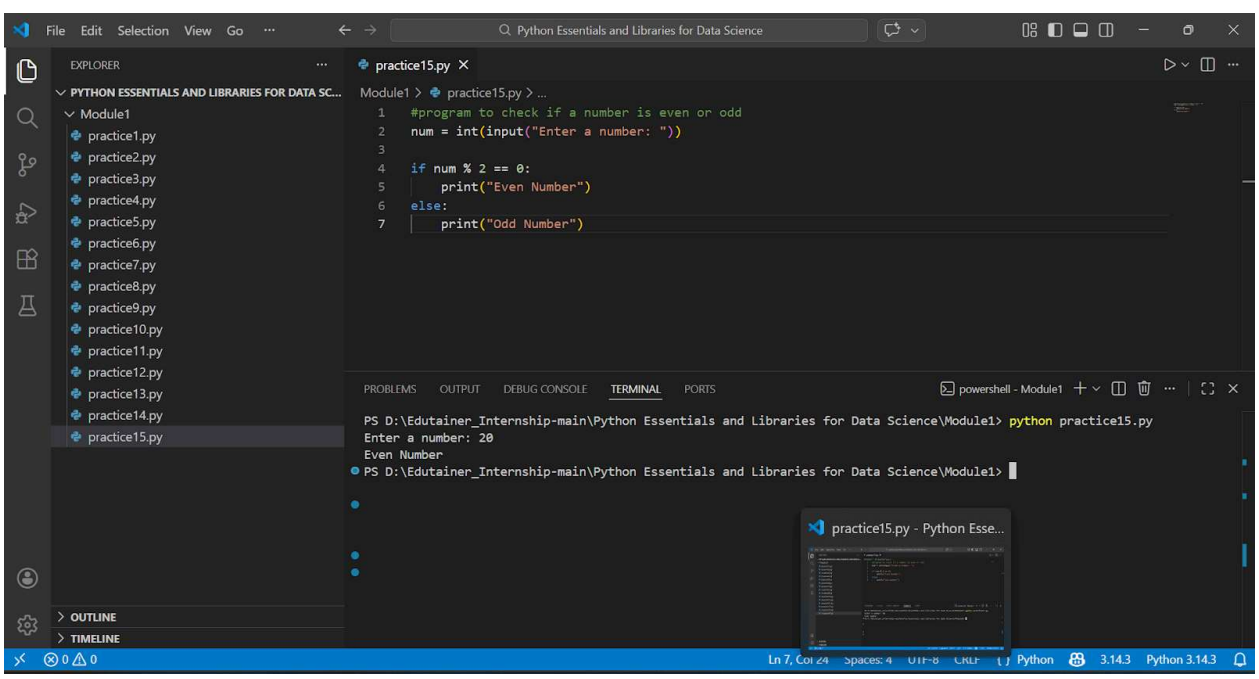
This program uses Arithmetic Operator (%).

% gives the remainder.

10 % 2 = 0

7 % 2 = 1

If answer is 0 its even else odd



**Conditional Statements:-**

Conditional statements are used to make decisions in programs.

**Python uses:**

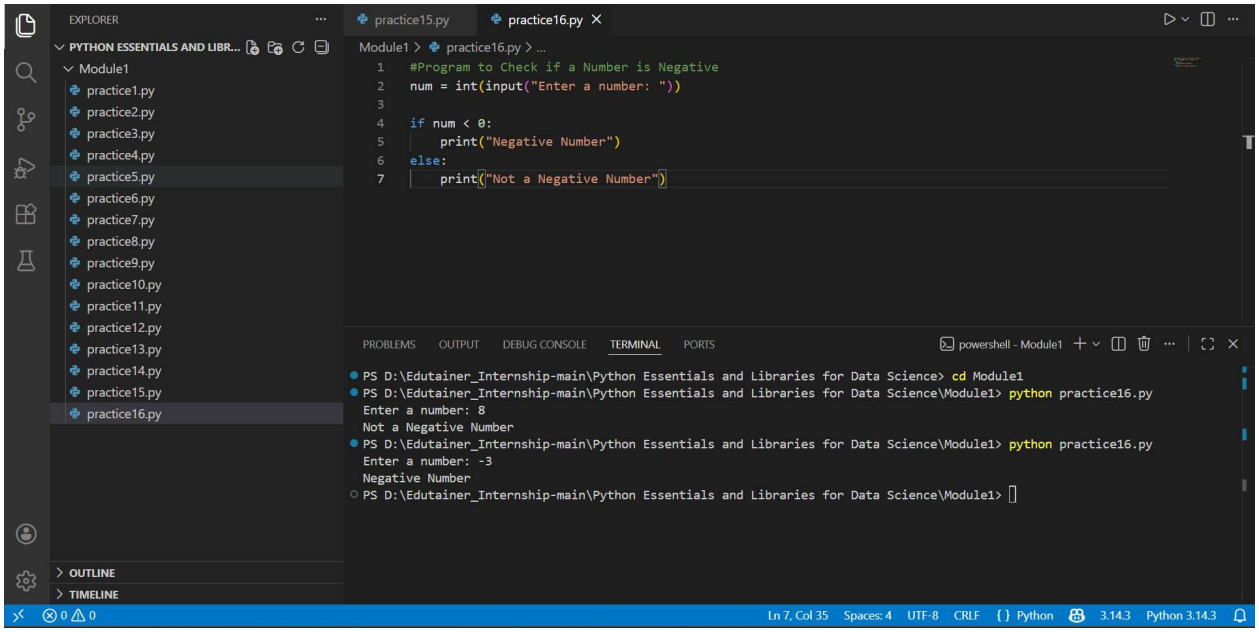
if
else
elif

Syntax:-

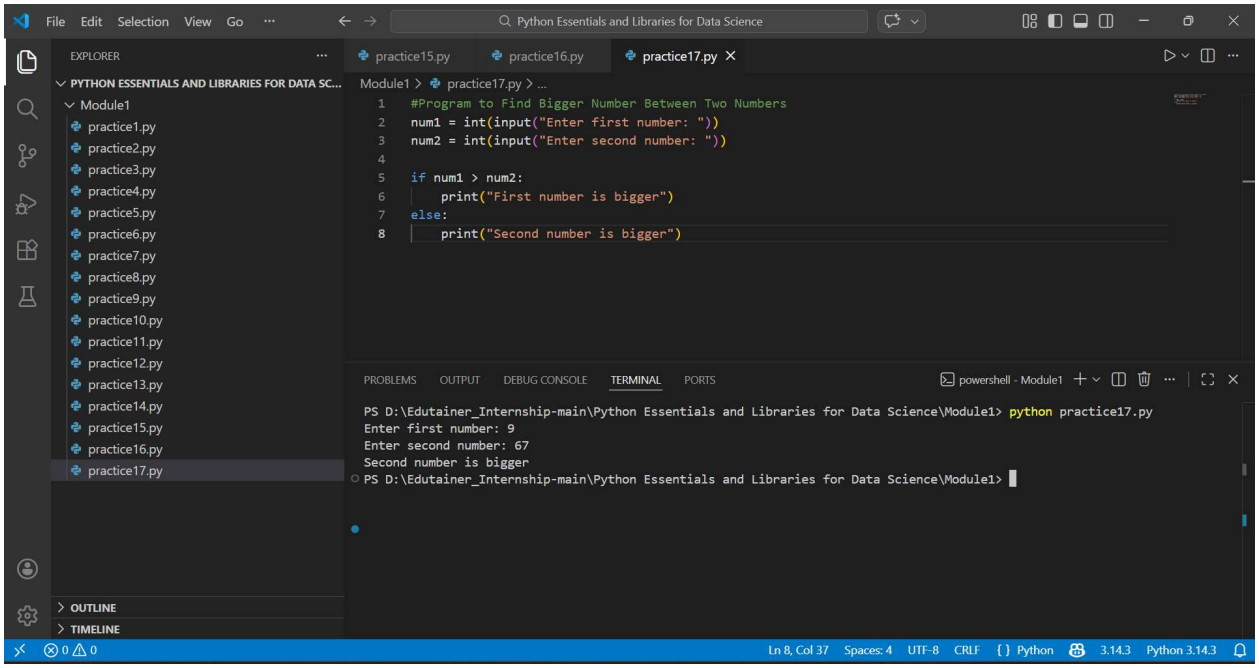
```
if condition:
    statement
else:
    statement
```

**Program to check if a number is negative**

If the number is less than 0 ,then its a negative number.




**Program to find bigger number between two numbers**


File Edit Selection View Go
Python Essentials and Libraries for Data Science
EXPLORER
practice15.py
practice16.py
practice17.py
PYTHON ESSENTIALS AND LIBRARIES FOR DATA SC...
Module1
practice1.py
practice2.py
practice3.py
practice4.py
practice5.py
practice6.py
practice7.py
practice8.py
practice9.py
practice10.py
practice11.py
practice12.py
practice13.py
practice14.py
practice15.py
practice16.py
practice17.py
Module1 > practice17.py > ...
#Program to Find Bigger Number Between Two Numbers
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
if num1 > num2:
print("First number is bigger")
else:
print("Second number is bigger")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - Module1
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module1> python practice17.py
Enter first number: 9
Enter second number: 67
Second number is bigger
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module1>
OUTLINE
TIMELINE
Ln 8, Col 37 Spaces: 4 UTF-8 CRLF { } Python 3.14.3 Python 3.14.3